instruction for drawing the cut and solving the calculations

1-The sliding is on the surface of a weak layer

2-start by drawing the sliding surface from getting the ε and then connect the slip surface and the horizontal surface to create the huge block then add up the areas

3-The angle of the back-face of the wall in the expression is α=0°,
the angle of wall friction" δ=φ

4-β (inclination of ground surface: β=5% to the left)

5-because of cohesion there might be a negative part in the stress diagram so consider only the positive area

the cut calculations

$\varphi_1 := 26\ \text{deg}$

$\gamma_1 := 20\ \frac{\text{kN}}{\text{m}^3}$

$C_1 := 5\ \text{kPa}$

$H := 12.77\ \text{m}$

from B-C (the slip surface)

$\varepsilon := 9\ \text{deg}$

$h_2 := \frac{H}{2} = 6.385\ \text{m}$

$\gamma_2 := 19\ \frac{\text{kN}}{\text{m}^3}$

$\varphi_2 := 30\ \text{deg}$

$c_2 := 15\ \text{kPa}$

$A := (93.642 + 59.45)\ \text{m}^2$

$L := 29.33\ \text{m}$

C.SL.d is the effective cohesion on the sliding surface

$C'_{SL.k} := 10\ \text{kPa}$

φ.csl.d is the effective angle of shearing resistance on the sliding surface

$\varphi_{csl.k} := 10\ \text{deg}$

constants

$\gamma_{c'} := 1.35$

$\gamma_{\varphi'} := 1.35$

Soil parameters	Symbol	Set	
		M1	M2 – Slopes [b]
Angle of shearing resistance [a]	$\gamma_{\varphi'}$	1,0	1,35
Effective cohesion	$\gamma_{c'}$	1,0	1,35
Undrained shear strength	γ_{cu}	1,0	1,5
Unconfined strength	γ_{qu}	1,0	1,5
Unit weight	γ_{γ}	1,0	1,0

1/ Active earth pressure coefficient:

$\delta := \varphi_1$ $\alpha := 0\ \text{deg}$ $\beta := 3\ \text{deg}$

$$K_a := \frac{\cos(\delta)\cdot\left(\cos\left(\varphi_1+\alpha\right)\right)^2}{\left(\cos(\alpha)\right)^2\cdot\left(\sqrt{\cos(\alpha-\delta)}+\sqrt{\frac{\sin\left(\varphi_1+\delta\right)\cdot\sin\left(\varphi_1-\beta\right)}{\cos(\alpha+\beta)}}\right)^2} = 0.3213$$

from A-B

$h_1 := \frac{H}{4} = 3.1925\ \text{m}$ $h := 4.39\ \text{m}$

$\sigma'_v := h\cdot\gamma_1 = 87.8\ \text{kPa}$

2/ The horizontal stress:

at A

$\sigma'_{h.a1} := -2\cdot C_1\cdot\sqrt{K_a} = -5.6681\ \text{kPa}$

when the horizontal stress diagrma will be zero since this is an active case with load on top

$Z_0 := 0.8821\ \text{m}$ you can get this from the graph or this equation

$\frac{x}{5.6681} = \frac{4.39-x}{22.54}$ $x := \frac{4.39\cdot 5.6681}{22.54+5.6681} = 0.8821$

at the end of h

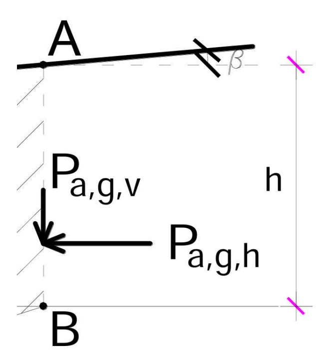
$\sigma'_{h.a} := \sigma'_v\cdot K_a - 2\cdot C_1\cdot\sqrt{K_a} = 22.54\ \text{kPa}$

3/ Then you calculate the Earth pressure force Pa,g,h from the positive area

$$P_{a.g.h} := \frac{1}{2}\cdot\left(K_a\cdot\gamma_1\cdot h - 2\cdot C_1\cdot\sqrt{K_a}\right)\cdot\left(h - Z_0\right) = 39.5341\ \frac{\text{kN}}{\text{m}}$$

4/ The vertical component of the active earth pressure is :

$$P_{a.g.v} := P_{a.g.h}\cdot\tan(\beta) = 2.0719\ \frac{\text{kN}}{\text{m}}$$



5/ The resultant of the active earth pressure force is then :

$$P_{a.g} := \sqrt{\left(P_{a.g.v}\right)^2 + \left(P_{a.g.h}\right)^2} = 39.5883\ \frac{\text{kN}}{\text{m}}$$

6/ This have to be divided into components normal (Pa,g,N) and parallel (Pa,g,T) to the slip plane: (ε represents the inclination of the sliding surface)

$$P_{a.g.N} := p_{a.g} \cdot \sin(\beta - \varepsilon) = -4.1381 \ \frac{\text{kN}}{\text{m}}$$

$$P_{a.g.T} := p_{a.g} \cdot \cos(\beta - \varepsilon) = 39.3715 \ \frac{\text{kN}}{\text{m}}$$

7/ Self weight of the sliding block :

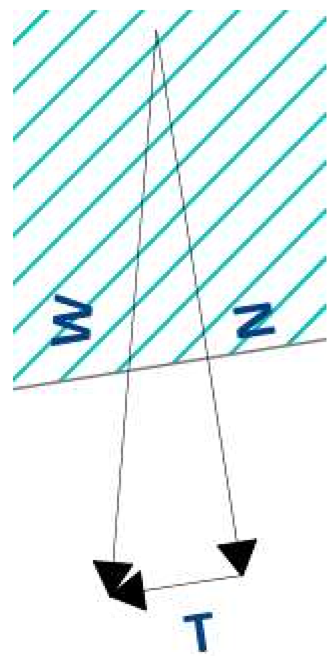
$$W := A \cdot \gamma_2 = 2908.748 \ \frac{\text{kN}}{\text{m}}$$

8/ Divided to components normal and parallel to the sliding surface

Force parallel to the sliding surface $T := W \cdot \sin(\varepsilon) = 455.0284 \ \frac{\text{kN}}{\text{m}}$

Forces normal to the sliding surface $N := W \cdot \cos(\varepsilon) = 2872.9365 \ \frac{\text{kN}}{\text{m}}$

Forces preventing the sliding :



Cohesion along the sliding surface

$$C_{SL.d} := \frac{C'_{SL.k}}{\gamma_{c'}} = 7.4074 \ \text{kPa}$$

$$C := C_{SL.d} \cdot L = 217.2593 \ \frac{\text{kN}}{\text{m}}$$

Friction from the self weight and friction angle

$$\varphi_{csl.d} := \text{atan}\left(\frac{\tan(\varphi_{csl.k})}{\gamma_{\varphi'}}\right) = 7.4414 \ \text{deg}$$

$$F := (N + P_{a.g.N}) \cdot \tan(\varphi_{csl.d}) = 374.7012 \ \frac{\text{kN}}{\text{m}}$$

Forces causing the sliding

$$T + P_{a.g.T} = 494.3999 \ \frac{\text{kN}}{\text{m}}$$

safety factor is adequate when it is more than 1

$$F_s := \frac{F + C}{T + P_{a.g.T}} = 1.1973$$